



Conflicting feelings: sociological and computational sentiment in workplace sentiment surveillance

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Abstract

Large employers are increasingly turning to sentiment analysis technologies as a form of corporate worker surveillance. Previous work has identified numerous problems with this trend, arguing that these technologies may inherit racial or gender biases from historical data, that the expansion of the surveillance apparatus itself threatens privacy and limits freedom of expression, and that the often proprietary algorithms may be inaccurate. This paper contributes to this conversation with a critical analysis of “sentiment” as it is operationalized by these technologies. As we argue, even on its own terms, sentiment monitoring software fails to capture the objective view of worker sentiment it imagines exists. Taking as a case study the sentiment analysis software platform Aware, we argue that these systems construct, through rhetoric and statistics, categories such as employee sentiment and toxic language that are divorced from the realities that these systems ultimately affect. In particular, we draw on Aware’s own public reports documenting its technology and experiments to reconstruct an analog of Aware’s sentiment analysis technology for study. We then compare the categories of “sentiment” and “toxicity” as they appear in the sentiment analysis literature, in Aware’s rhetoric, and in the analyzed dataset. Through this combination of technical and critical analyses, we place contemporary sentiment analysis technologies within a troubled historical context of workplace sentiment surveillance and illustrate a method of analyzing contemporary algorithmic technologies in terms of their underlying critical concepts that may have applications beyond critiques of workplace sentiment analysis.

Keywords Algorithm auditing · Sentiment analysis · Bossware · Large language models

1 Introduction

According to a recent report by employee management software vendor Aware, the cause of the crushing fatigue and burnout experienced by frontline retail employees is not difficult working conditions, such as long hours, job insecurity, the lack of benefits, and low pay—it is Karens (Aware 2023), a derogatory term used to refer to white entitled women (Brady et al. 2023). Aware sells a software platform that, among other things, promises managers the ability to reveal employees’ sentiments—their frustrations, their concerns, the things that burn them out, and even their intentions to

sell company secrets—by analyzing their emails and social media profiles. Such employee monitoring software, colloquially known as “bossware,” has become increasingly ubiquitous in a post-pandemic world of remote work and ever expanding global workforces (Ball 2022; Munn 2024). It is therefore imperative that we understand their limitations and potential harms.

Existing scholarship on such workplace surveillance systems recognizes a sometimes delicate tension between protecting employees from harm and harassment and invading employees’ privacy or infringing on their civil liberties (Abraham et al. 2019; Ball 2010; Holland and Tham 2022; Kayas 2023; Sewell and Barker 2006). Complicating this tension is the growing concern that algorithmic technologies may perpetuate past discrimination and injustices reflected in the historical data on which they are often trained (Benjamin 2019; Chun 2021; Eubanks 2018; Noble 2018 O’Neil 2017).

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New fields such as algorithm auditing (Ajunwa 2019; International Panel on the Information Environment 2024a; Metaxa et al. 2021) and algorithmic accountability (AI Now Institute 2023) have developed techniques for holding algorithmic systems accountable to relevant regulatory frameworks. However, although third-party algorithm auditors often gain privileged access to proprietary data and algorithms, they are often legally constrained to assessing differential impacts on protected demographic groups (Ajunwa 2019; Gilbert et al. 2023). Such restrictions prevent these approaches from addressing deeper theoretical and epistemological difficulties with the way that algorithmic employee monitoring systems categorize employee thought and action (Kaufmann et al. 2019; Ugwuodike 2022). These restrictions have also led to the strategic use by companies of audits limited in scope to lend credibility to products that may nevertheless have risks that fall outside the scope of algorithmic audits in a process known as “audit washing” (Goodman and Trehu 2022). While there has been mounting pressure on companies such as Aware to make public additional details about models and data that may enable more sophisticated oversight in the future, access to such information remains limited at present (International Panel on the Information Environment 2024b).

In this paper, we examine the conceptual underpinnings of the application of computational sentiment analysis techniques to employee monitoring. We adopt, as a case study, a report by Ohio-based bossware vendor Aware that makes public details of an experiment conducted using their platform (Aware 2023). Because the details of Aware’s platform and experiment made public in the report are somewhat limited, it is necessary to exercise caution when making strong claims about the Aware platform on the basis of this report. Nevertheless, due to the still uncommon level of detail in the report for companies in this space, we present this study as a preliminary investigation into the types of questions and approaches that may prove helpful for exercising greater oversight of such technologies in a future where model and experimental detail are more readily available. To that end, we contend that this study can inform efforts to determine what types of details may be most beneficial to request from such companies.

The report, which claims to identify causes of frontline retail employee burnout, uses publicly available data drawn from Reddit, and so offers a unique opportunity to partially replicate an experiment conducted on an otherwise proprietary platform. By tracing the genealogy of “sentiment” as it appears in applied sociological research and in natural language processing, we argue that bossware companies such as Aware use the term in two mutually incompatible ways, raising questions about the application of sentiment technologies to the analysis of employee sentiments. We then augment this genealogy with a data scientific investigation of

the Reddit data and a variety of sentiment analysis systems, which we use as proxies for the Aware platform. We conclude that computational sentiment analysis technologies, applied to workplace communications, represent a categorical mismatch that risks constructing categories of sentiment that misrepresent employee behavior and intent, with the potential of causing real-world harm if used as the basis of managerial decision-making.

This paper is divided into two main sections. In Sect. 2, we examine the histories of workplace sentiment analysis and computational sentiment analysis, arguing that these histories represent divergent conceptions of “sentiment,” and calling into question the application of computational sentiment analysis to workplace communications. In Sect. 3, we analyze the technical details made public in the Aware reports and partially reproduce elements of the Aware system on the basis of these details to argue that, in practice as well as in theory, computational sentiment analysis poses a realistic risk of misrepresenting workplace sentiment, limiting its utility and introducing potential real-world harms.

2 A critical analysis of sentiment surveillance

2.1 Sentiment & aware

We focus, in this paper, on the limits and risks of sentiment analysis techniques for analyzing digital employee communications. Founded in 2017, Aware develops an enterprise business intelligence platform and counts as customers a number of major American and international corporations, including Walmart, Starbucks, Delta, and AstraZeneca (Field 2024). The Aware platform provides a number of services beyond sentiment analysis, and only some of these corporations have confirmed usage of Aware’s sentiment analysis technologies. Aware was acquired by “Human Risk Management” firm Mimecast in 2024, as we were conducting this study (Mimecast 2024) and it is unclear how the two companies will integrate. Nevertheless, given the size of the workforces that Aware’s original clients represent, even a selective use of sentiment analysis has the potential to impact a large number of workers, underscoring the need to consider carefully the potential harms of these technologies.

In this section, we tease apart two distinct senses of the term “sentiment” as it appears in Aware’s marketing materials, and especially in a report made public on Aware’s website (Aware 2023; Schumann 2023). The first sense is as an appeal to the perennial concerns of the managers that constitute Aware’s customers. Aware promises managers the ability to measure—and therefore manage—employee sentiment. With Aware’s technology, managers can identify aspects of the workplace environment causing resentment,

drop in morale, or burn out among employees to better support them. In short, Aware claims to reveal the authentic “Voice of the Employee” (Deloitte and Aware 2021). Additionally, Aware advertises the ability to perform toxicity detection, a variety of sentiment analysis that Aware defines as assisting in locating “hard evidence” of individual employee misbehavior, including, “harassment, discrimination, microaggressions,” and other behaviors that contribute to, “creating a toxic working atmosphere” (Aware 2023; Deloitte and Aware 2021, 4). Toxicity, Aware claims, is, “associated with reduced morale, performance decline, high turnover, and worsening brand reputation” (Aware 2023).

This use of sentiment as a key managerial category is common among other bossware vendors with similar sentiment analysis product offerings. Happiness Index, for example, although exchanging the language of “sentiment” for a neuroscientifically inflected “happiness,” nevertheless articulates a similar focus on measuring the inner affective state of employees through an algorithmic analysis of their digital communications. Terminological differences aside, both platforms arrive at a similar destination, with happiness metrics and an “Employee Voice” product promising insight into the invisible sentiments that drive workplace efficiency (Whitehead-Smith 2024). Qualtrics, likewise, is a bossware vendor that leverages sentiment analysis techniques to track employee communications across internal platforms and social media (Qualtrics, n.d.). Like the other two vendors, Qualtrics’ marketing copy foregrounds the revelation of employee “intent” and the centering of employee “voice” through sentiment analysis technologies.

The second sense of sentiment, common to all three of these companies, refers to the natural language processing sentiment analysis technologies that underwrite their respective claims to reveal employee frustrations, happiness, intent, and voice. All three treat sentiment analysis techniques as transparent lenses through which to view the employee sentiment on which workplace culture and productivity depend. Yet, as we show in the remainder of Sect. 2, these two senses of sentiment possess distinct, albeit interconnected, genealogies that complicate the otherwise seemingly intuitive proposition of using sentiment analysis techniques to measure “sentiment.”

2.2 Sentiment in the workplace

Ball (2010) places bossware within a history of workplace surveillance practices stemming from twentieth-century management theories advocating careful measurement of worker productivity. This history has, in turn, traced a course from management of the collective to, increasingly, the “mastering of the individual worker” and the consequent “uninterrupted monitoring of workers’ lives” (Ajunwa et al. 2017, 772). Given Aware’s focus on the proposition that

there exists a measurable quantity—sentiment—and that measurements of this quantity can aid in managing employee wellness, productivity, and insider threats, we view Aware’s use of sentiment analysis as existing within a long history of twentieth-century sociological theory responsible for developing the concept of sentiment as a tool for industrial and civic population management (Albrecht and Chun 2024; Roethlisberger and Dickson 1939; Whitehead 1938). This section places Aware’s rhetoric at the end of this history and, in doing so, illuminates the assumptions and instabilities with which that concept has been invested by this history. The question then taken up in the remainder of the paper, then, will be: does Aware’s use of sentiment analysis technologies manage to escape this history, or does it reproduce the same issues that have long haunted sentiment as a tool of surveillance and management?

Per Albrecht and Chun (2024), one of the earliest examples of managerial sentiment analysis that established the tradition within which Aware and its competitors exist was a series of experiments conducted from 1927 to 1932 at the Hawthorne Works, a Western Electric manufacturing plant. In collaboration with Harvard Business School, researchers separated out six women from a group responsible for assembling electrical relays and modified experimental variables such as break times and work hours in search of peak worker productivity. When changes to environment and schedule did not produce predictable results, researchers realized that the women’s sentiments toward the changes being made and toward their status as participants in these special experiments mattered more to their productivity than their working conditions per se (Roethlisberger 1962). This discovery led to the researchers interviewing almost all workers in the factory and producing tables that classified their responses to certain topics as negative or positive. Although they initially simply summarized these responses, they quickly moved to transcribing entire interviews to delve more deeply into worker sentiment. These researchers, however, also underscored the limitations of their studies—in particular the coarseness of their word-sentiment tables (Whitehead 1938).

This study would deeply inform the development of management theory, lending its name to the subsequently formulated “Hawthorne effect,” which holds that individuals modify their behavior in response to the knowledge that they are being observed (French 1953; Jones, Stephen 1992; McCambridge et al. 2014). Sentiment analysis as a way to “understand” workers’ sentiments and the effects of those sentiments on cooperation with management took a seemingly darker turn when it was deployed by Alexander Leighton’s Bureau of Sociological Research (BSR) to inform the management of Japanese internment camps during World War II. Building on the results of the Hawthorne study, the BSR put classic anthropological methods of analysis toward

the end of effective administration of the camps (Chun et al. 2024). Using methods such as quantitative measurements of satisfaction and dissatisfaction, the BSR scaled the sentiment analysis methods of Hawthorne Works to the larger population of hostile internees, who were forcibly removed from their homes in the Western US and interned regardless of citizenship status.

Fundamental to the BSR's approach was the wager that administration guided by sentiment analysis could lead to greater cooperation with camp management at a time when even those who were initially sympathetic to the US cause had called a general strike due to poor working and living conditions. The BSR's stated goals of democratizing camp governance by giving internees a voice—albeit one revealed through statistical methods—and of mitigating the risk of insider sabotage that so concerned military administrators find echoes in the rhetoric of modern bossware companies. Notably absent from this modern rhetoric, however, are the earlier studies' reflections on the limitations of sentiment analysis, and in particular the coarseness of these positive versus negative evaluations and difficulties in understanding subtleties and context (Chun et al. 2024; Roethlisberger and Dickson 1939; Whitehead 1938). As Chun (2024) argues, the BSR's archives evince many of the same difficulties that plagued the Hawthorne study, namely, the sensitivity of sentiment to the subjects' awareness of the conditions of their own observation, and the insensitivity of the methods of the social scientists to the situational ironies and resistances expressed by the subjects of their studies.

2.3 Quantifying sentiment

Although Aware, through its rhetoric, places itself within the historical genealogy of managerial sentiment analysis, the step from the quantitative population statistics of the BSR to the machine learning methods offered by bossware suppliers is not necessarily the natural result of increasing computational power. Although linked by shared historical preoccupations with consumer sentiment and opinion polling, computational sentiment analysis and applied sociological studies of worker sentiment trace two distinct historical arcs before their reunion in the rhetoric of contemporary employee surveillance. This section teases apart the variegated notions of “sentiment” at work in bossware discourse and exposes a fissure between the computational sentiment analysis literature and the claims made by bossware firms.

One of the earliest instances of computational “sentiment analysis” per se was a 2001 effort to extract market “sentiment” from the text of online message boards discussing specific stocks (Das and Chen 2001; Feldman 2013). “Sentiment” in this case refers to investor expectations of future market movements, as opposed to the intangible personal affects that were the subject of the Hawthorne study,

although certainly investors' personal sentiments may at times be tied to those of the market. From there and throughout the 2000s and 2010s, “sentiment analysis” came to be used more broadly across a range of application domains to refer to methods for summarizing and quantifying subjective orientations expressed especially through text. This wider circulation of the term “sentiment” was due in part to the burgeoning range of varieties of online text amenable to computational analysis, which tracked the increasing adoption and commercialization of the world wide web (Pang et al. 2002). This circulation in turn diluted the original etymology of the term “sentiment” in market analysis and gave rise to competing terms such as “opinion mining.” (Liu 2012). The uncertainty in the terminology unifying so many disparate studies mirrored an underlying ambiguity in the scientific quantity nominally under study.

Methodologically, this early work drew on content analysis research in Communication and Media Studies in framing its task as an attempt to quantify the positivity or negativity, broadly construed, of a passage of text (Dave, Lawrence, and Pennock 2003; Krippendorff 1980; Osgood et al. 1957; Stone et al. 1966; Turney 2002). This work often assigned each word a particular *valence*, or a measure of the word's semantic polarity and magnitude (Taboada 2016). For instance, the word “dislike” might have a negative polarity of small magnitude, whereas “despise” might have a negative polarity of greater magnitude and “like” a positive polarity. The sentiment of an entire text, then, became a function of the polarities and magnitudes of its constituent words.

These sentiment scores were then turned toward assessing the sentiments of the online reviews of books, movies, and consumer products that populated the early web. These new media forms offered a tempting target for sentiment analysis research in that they presented researchers with a question that naturally lent itself to the type of quantification required by sentiment analysis technologies. Web reviews often included a star rating or other numeric score alongside a textual comment that gave the reviewer the opportunity to contextualize their score. Given a restaurant review with a four-star rating and a comment praising the food but criticizing the customer service, researchers could ground the otherwise nebulous question of quantifying sentiment by replacing it with a more easily measurable question of whether it was possible to predict the star rating from the text (Lei and Liu 2021; Mäntylä et al. 2018). The task of research, then, was to refine the computational methods used to analyze the text and bring their predicted star ratings ever closer to the ratings found in the actual data.

By formalizing the problem in this way, researchers were able to extend sentiment analysis methods to other domains. Examples of previous research, each adapting the “positive” and “negative” categories of sentiment analysis subtly to fit a diverse range of media, involved classifying good or bad

news headlines (Balahur et al. 2013; Ku, Liang, and Chen 2006); pros and cons of a product (Kim and Hovy 2006); candidate likely or unlikely to win an election (Kim and Hovy 2007; Mohammad et al. 2015); support or opposition for proposed legislation (Bansal, Cardie, and Lee 2008); depression or not from social media postings (Babu and Kanaga 2022); or, in a return to the field's origins, stock price likely to go up or down (Bouktif et al. 2020). Over 20 years, the term sentiment migrated from an analysis of markets to revealing the interiority of the human actors from which those markets are constituted.

More recently, applications of sentiment technologies to social media content moderation have fallen under the emerging subfield of sentiment analysis known as toxicity detection (Risch and Krestel 2020). Rather than revealing opinions or sentiments, toxicity detection focuses on toxic, abusive, or hateful behavior on social media platforms, as identified by high negativity, possibly with the addition of profanity, slurs, and similar language. Toxicity is defined as online content that is considered derogatory, abusive, or insulting, and which may have offline consequences, such as denigration of a person or group (Pachinger et al. 2023; Talat et al. 2017). Lacking star ratings or other obvious numeric correlates with toxicity, toxicity detection datasets are often produced through manual annotation of social media comments developed specifically for this type of research. Because these annotations are developed within the context of the research rather than encountered within the broader context of online behavior, such as reviewing, researchers have analyzed the annotation process itself and discovered it to be extremely context sensitive, with judgements about what counts as toxicity depending heavily on the experimental design (Pavlopoulos et al. 2020).

To our knowledge, sentiment analysis research has never been put, in the literature, to the use of assessing workplace morale, burnout, or criminal intent, per Aware's marketing materials and the history of workplace sentiment with which they are in dialog. Moreover, as has long been recognized, the data used to train sentiment analysis systems necessarily encode sociolinguistic assumptions concerning speakers' economic, geographic, and educational status (Henrich et al. 2010), and therefore tend to bias the interpretation of the analysis along axes, such as age (Diaz et al. 2018), gender (Thelwall 2018), disability (Narayanan Venkit, Srinath, and Wilson 2023), and combinations thereof (Kiritchenko and Mohammad 2018). Sentiment analysis technologies likewise often overlook sexist or misogynistic content on the basis of such encoded assumptions (Adams 2023). As sentiment analysis technologies have proliferated throughout industry and government, these embedded assumptions have translated into real-world harms, such as gender discrimination on the basis of historical hiring patterns (Winick 2018) and circumvention of labor protections (Newman 2017). The

question, then, turning to Aware's findings, is how Aware justifies the application of its sentiment analysis technology to the novel domain of workplace sentiment and whether it manages to avoid the possible harms that may stem from misinterpretations of employee data by algorithmic analysis. As Taylor (2023) argues, "sentiments, or how real people actually feel, rarely map rationally onto statistics" (p. 11).

3 Replicating Aware's findings

The challenge facing bossware companies such as Aware, as argued in Sect. 2, is that the workplace sentiment at the center of the managerial imaginary possesses a genealogy that diverges from that of the collection of topics of study that fall under the rubric of computational sentiment analysis. Applying an off the shelf sentiment analysis system directly to internal employee communications is a problem that has not been well studied in the scientific literature. Of course, bossware systems are, as a rule, proprietary, and it is therefore impossible to know what contributions companies may have made to the current state of the art in sentiment analysis to bridge this gap, and how they may have evaluated the efficacy of their technologies. This section, therefore, draws on details Aware has made public about its system to investigate the theoretical possibilities and limitations of a sentiment analysis system conforming to those public specifications. Our study constitutes not a full replication, but a reconstruction of what is possible in bossware, and a questioning of the assumptions behind such technology.

The basis for our partial reconstruction of Aware's technology lies in a collection of documents Aware published to their website comparing their system and data to publicly available language models and datasets. Although we lack direct access to the Aware platform, we do have access to the open-source Llama-2 language model, which Aware used as a benchmark for its technology, and publicly available Reddit data, which Aware identifies as within the scope of its platform. Using these public models and data as proxies, we attempt, to the extent possible, to replicate Aware's results concerning employees' relationship to management and to Karens. In doing so, we gain insight into Aware's operationalization of sentiment and the interpretive challenges that accompany the algorithmic assessment of workers' interior states.

3.1 Data

Aware's first report describes a study of frontline retail workers and identifies interactions with "Karen" customers as the most common cause of burnout and turnover (Aware 2023). The study analyzed 152,716 Reddit comments from the public subreddits of eight major retail and frontline employers.

Table 1 Comment totals of eight top retailer subreddits

Subreddit	Comments
BestBuyWorkers	412
DollarGeneralWorkers	2309
MichaelsEmployees	17,907
WalmartEmployees	1413
TjMaxx	7490
Sephora	122,752
FootLocker	12
AmericanEagle	421
Total	152,716

^aAfter experimenting with several prompts, we report the values for “Classify this message as either ‘Toxic’ or ‘Healthy’, with no explanation—just the category as a one word response.” and “Classify this message sentiment as either ‘Positive’ or ‘Negative’ or ‘Neutral’, with no explanation—just the category as a one word response.” respectively

Reddit is a public Internet forum, and so, the nature of communication between employees and between anonymous members of the public will necessarily differ from that of internal communication platforms. Aware acknowledges this distinction, hypothesizing that anonymous Internet forums may have higher toxicity than workplace communications. However, operating on the assumption that the half of new frontline hires who do not quit within the first 120 days will, “turn to coworker forums like Reddit or Workplace from Meta to find the support and empathy their leaders fail to provide,” Aware argues that it is incumbent on employers to “look beyond what their people say at work” to form an accurate picture of employee sentiments (Aware 2023). Following Aware’s justification, we also adopt Reddit as an appropriate dataset for examining the limitations of employee sentiment analysis.

Using the numbers provided by Aware’s report, we were able to replicate the study’s dataset with reasonable confidence. The report describes the overall Reddit dataset as being composed of 152,716 posts gathered from the subreddits of 8 unnamed major retailers during the period from January of 2023 to April of the same year. Correspondingly, we gathered Reddit posts from this time period using Arctic Shift (Heitmann 2023), an open-source tool with pre-scraped snapshots capturing the entirety of Reddit’s data, thereby ensuring that replication is feasible using publicly archived data, and filtered them by the National Retail Federation’s list of the top 100 retailers of 2023 (National Retail Federation 2023). Using this approach, we located eight subreddits with post counts that summed to exactly 152,716 during

this time period. We then used this collection of posts as our primary dataset for subsequent experiments (Table 1).¹

Despite the precise comment totals, a number of the report’s other examples and figures differed from our own as measured on this dataset, suggesting that Aware’s exploratory analysis may have been more far reaching than their final reporting. Among the example comments cited in the report, for example, was a comment from the */r/starbucks* subreddit, which alone contained more than 164,000 comments—over 11,000 more than were reported for the entire final sample. Likewise, the report included the statistic that the most active subreddit in their sample averaged 1553 comments per day, implying a total of over 186,000 comments—over 33,000 more than reported for the entire sample.

Filtering the above dataset for the word “Karen” and manually reviewing the results, we were able to extract 64 posts relevant to the Karen topic from our sample, accounting for 0.04% of the total dataset—fewer than the reported 0.1%, or approximately 153 expected posts cited in the study, although potentially within expectations depending on the degree of precision used in the reporting. Taken together, the precise comment totals from the employee subreddits coupled with the roughly similar magnitudes of comments in the Karen subsample suggest that we were able to recover a reasonable representation of the data used in the Aware study. Naturally, Karen-like behavior may have been reported in the posts without necessarily mentioning the keyword Karen. Given that Aware used the keyword, our results compare to theirs.

3.2 Models

Producing an adequate model for comparison is more difficult as, unlike the Reddit data used in the report, Aware’s sentiment analysis system is not publicly available. Nevertheless, in their reports, Aware’s data scientists draw their own comparisons to Meta’s open-source Llama-2 13b model (Touvron et al. 2023). Partly on that basis, we argue that an analysis of Llama-2 and comparable sentiment analysis systems can offer insight into Aware’s broad conceptualization of sentiment.

¹ There were several combinations of subreddits that summed to this total. While most differed only by very small subreddits that are unlikely to significantly affect the final result, several did differ to more significant degrees. We chose the combination with the most employee-oriented subreddits (as opposed to general brand subreddits), which also seemed roughly to accord with other statistics presented in the report. Further work might profitably pursue the following analyses in parallel across all possible combinations and synthesize the results of doing so.

Table 2 Sentiment and health scores for four sentences

Sentence	Sentiment		Health	
	Aware	Vader	Aware	Detoxify
1) That pin is so gorgeous	0.85	0.88	0.96	0.86
2) That's my sarcastic way of saying whoever did this was f**cking amazing bc WOW	0.84	0.92	0.33	0.02
3) I'm so sad: I hadn't seen them yet and this is not what I was expecting	0.17	0.12	0.97	1.00
4) What a crock of sh*t. Your store manager sounds like a jerk	0.15	0.23	0.16	0.01

We report the numeric sentiment and health scores produced by Vader and Detoxify, respectively. Our other models produced classification labels, but we did not attempt to use them to generate numeric scores

To demonstrate the efficacy of their system, Aware documents a benchmark test comparing the Aware platform to the base Llama-2 model on the task of zero-shot sentiment analysis (Schumann 2023). Aware reported an 87.3% accuracy for their own models versus a 62.7% accuracy for Llama-2. Although they do not specify the dataset used in the comparison, the figures suggest that the models must have agreed on at least 50% of the examples.² Without knowing what dataset Aware used for the experiment, it is impossible to comment on potential issues such as definitions of toxicity, as discussed in Sect. 2.3. Moreover, accuracy scores themselves can be misleading without knowing the distribution of categories in the data. The Karen study, for example, reports that the Reddit data are 86% neutral, as estimated by the Aware system. If this were the case, a naive classifier that completely ignored the data and simply predicted "neutral" for every item would therefore also report 86% accuracy.³ That said, Aware's researchers treat the Llama-2 system as a potential competitor, and so, we adopt this stance as the basis for our comparison as well.

Like other modern LLMs, Llama-2's ability to perform sentiment and toxicity analysis is due largely to specialized training such LLMs receive across a range of common NLP tasks (Touvron et al. 2023). Because both sentiment and toxicity analysis are common concerns within the broader NLP literature, prompts and datasets related to both sentiment and toxicity were a part of Llama-2's development and training. Our usage of the same default Llama-2 model, then, should produce similar results to those observed during Aware's experiments, as the report notes they did no additional training during the comparison. The one key missing piece of information is how Aware prompted Llama-2 to perform sentiment analysis.

Llama-2's ability to conduct sentiment analysis hinges on its training, its performance will tend to be highest when the prompt resembles that seen during training. Lacking access to Aware's prompt and to the specific prompts used during training, we modeled our prompt on the documentation provided by the developer of the library we used to run Llama-2, HuggingFace (HuggingFace 2024). The prompts for both sentiment and toxicity were as follows:

```
[INST] <<SYS>>\nClassify the sentiment of the text
into neutral, negative or positive. \n<</SYS>>\n\
nText: reddit-comment\n [/INST] \n\nSure! Sentiment:
\n\n
```

```
[INST] <<SYS>>\nClassify the toxicity of the text
into toxic or healthy. \n<</SYS>>\n\
nText: reddit-comment\n [/INST] \n\nSure! Toxicity: \n\n
```

The core of the prompt is the instruction: "classify the sentiment of the text into neutral, negative or positive." Markup such as "[INST]" and "<<SYS>>" is parsed specially by the library and is used by the model to understand the structure of the prompt. Its inclusion noticeably improved the consistency of our results. The final segment of each prompt, "Sure! Sentiment:" represents the beginning of the model's answer. Models such as Llama-2 work by completing an initial passage of text by predicting the next words. Interestingly, because much of the reason models are trained to recognize toxicity is so that they can monitor their own output and prevent themselves from generating toxic speech, we found that Llama-2 frequently refused to classify the sentiment of our data due to its refusal to handle toxic content (Röttger et al. 2023). By prompting the model to begin its answer with "Sure!", we were able to elicit its toxicity classifications more reliably. These prompts yielded categorical classifications of sentiment and toxicity, rather than scores, and so were used in comparisons with Aware's summary statistics of the sentiment and toxicity of the entire dataset (Table 2).

Additionally, because Aware's report seemed to indicate that Llama-2 was being used in a fashion deeply rooted in the contemporary sentiment analysis literature, we also

² This assumes, in the worst case, that Llama-2 correctly classified the 12.7% missed by Aware and then the remaining 50% that Aware also correctly classified.

³ Lacking a more specific definition in the report, we adopt the conventional definition of classification accuracy as correctly classified cases as a proportion of all cases.

employed several other common sentiment analysis systems to help hedge against possible idiosyncrasies of our Llama-2 usage. All told, we made use of three additional sentiment and toxicity systems: Vader, an open-source rule-based sentiment analysis system widely used in academia and industry (Hutto and Gilbert 2014); Detoxify, an open-source fine-tuned transformer toxicity classifier (Hanu and Unitary 2020); and the closed-source GPT-3.5-Turbo API (OpenAI 2024).

We found that our models performed comparably to the Aware system on several points and differed in others. Our first point of comparison was a table of four example sentences and associated sentiment and toxicity scores, reproduced alongside Vader and Detoxify in Table 2. While four sentences do not reach statistical significance, they nevertheless provide a useful qualitative point of comparison.

As a starting point, because these examples demonstrate the system working as intended, they offer perhaps the most natural route to understanding how Aware construes sentiment and toxicity in its analyses. To better understand the factors that contributed to these scores, we re-ran our models on versions of these sentences edited, so that high polarity words and clauses were replaced by neutral stand-ins or omitted entirely. The sentence 1 became neutral, as expected, when “gorgeous” was replaced by a neutral adjective, such as “small.” Sentence 2 became healthy when “f*cking” was removed.⁴ Sentence 3 became neutral when “I’m so sad” was removed. Sentence 4 became neutral and healthy when both “crock of sh*t” and “jerk” were removed or replaced.

These results are not surprising for off the shelf sentiment analysis systems trained on a general range of web data. It is less clear, however, whether they lend themselves to the analysis of employee morale at the center of Aware’s value proposition. Sentence 2, “That’s my sarcastic way of saying whoever did this was f*cking amazing,” appears to express a positive sentiment, and the notion that obscenity, as detected by a sentiment analysis system, contributes to a toxic workplace would seem to represent a fairly strong value judgment on Aware’s part. In its more complete context in the data, “that” in this comment refers to the poster’s previous comment, “I can draw stick figures lmao.” This remark, drawn from the Starbucks subreddit, expresses the poster’s amazement in response to a photograph of a colleague’s highly skilled chalk artwork of a unicorn, advertising the introduction of the Unicorn Frappuccino. It is worth asking whether obscenity, which often forms the basis of toxicity classification datasets for anonymous online

Table 3 Classification percentages on full dataset

Sentiment	Aware	Vader	Detoxify	GPT-4o-mini ^a
Positive	8%	63.62%		42.04%
Neutral	86%	19.12%		33.87%
Negative	8%	17.26%		24.09%
Toxicity				
Healthy	89%		97.6%	91.71%
Toxic	11%		2.4%	8.29%

We use a 0.5 probability threshold for classification using Detoxify. Vader distinguishes polarity with positive and negative values. All other systems produce categorical classifications directly

Note that although the toxicity percentages are similar, these statistics do not guarantee that the systems agree on which posts are toxic. Without the precise results produced by Aware, it is difficult to be more precise at this level of granularity, although it is reasonable to suspect the sets of classified comments are not totally disjoint. Still, insofar as the systems do agree on some subset of toxic comments, it may be useful to examine the difficulties facing the off the shelf systems, as these may surface issues and place a burden of proof on Aware to demonstrate that their system avoids them

platforms, creates a hostile workplace environment (Dynel 2016) Table 3

Running our models on the full dataset yields comparable results for toxicity, although Aware is considerably more neutral in its analysis of sentiment.⁵

3.3 Results

The Aware report describes an experiment using the Reddit data through which they assess causes of employee burnout. It must be stated that this report is not a scientific study, although it is framed through the language of data science. The clear intent of this document is to clarify the value proposition of Aware’s software and ultimately to pitch a product to potential customers in charge of managing large workforces. In following the form of data science research, however, the report does offer some insight into the operations and analytic capabilities of the system. In this section, we therefore take Aware at its word in examining this experiment as data science. Our goal is to evaluate whether the conceptual gap between competing notions of sentiment present problems in practice for platforms such as Aware.

Among the day-to-day stressors that Aware identifies as causes of retail employee burnout, the foremost factor was employees’ customer service interactions with customers identified in the report as “Karens.” The term “Karen,” which Aware defines as “a customer who escalates complaints beyond what is considered reasonable,” rose to

⁴ We ran these experiments with the original, uncensored language drawn from the Reddit data. The double asterisks in Aware’s table appears to be a typo.

⁵ Llama-2 was omitted from this experiment due to limited computational resources available to run it on the full dataset.

prominence during the pandemic and has since become an object of academic interest as well as broad popular recognition (Aware 2023; Bhasin et al. 2020; Brady et al. 2023; Dubrofsky 2022). Aware argues that the value of their platform lies in its ability to surface issues such as stressful dealings with Karens, so that management can act to better support employees and reduce burnout. In this section, we investigate these findings and demonstrate ambiguities in Aware's results that complicate their interpretation.

Aware states that, although only 0.1% of all the messages studied mentioned Karens, those messages were 36% more negative and 100% more toxic than the average post in the dataset, yielding a predicted overall drop in workplace sentiment of 3% (Aware 2023). The report further elaborates on these figures using a table reproduced here as Table 4:

The 36% increase in negativity cited in the report seems to refer to the multiplicative 37.5% increase from the 8% negativity of the average employee Reddit post to the 11% negativity of posts belonging to the Karen topic. Likewise, the doubling of toxicity likely refers to the 82% increase from 11 toxic to 20% toxic in the Karen topic. The 3% drop in overall workplace sentiment due to the inclusion of the Karen topic, by contrast, seems to refer to the 3% additive difference between the 11% negativity of the Karen subtopic and the 8% negativity of the overall sample. However, there are several complications in interpreting these figures arising from inconsistencies in the table and treatment of the claims concerning it.

Part of the difficulty with interpreting the 3% difference in negativity as a drop in "overall workplace sentiment" is that the phrasing of the report seems to imply that the inclusion of the Karen topic in the overall results raises the average negativity by 3%. This suggests that Karens are to blame for

a 3% reduction in workplace efficiency as measured in terms of the managerial category of worker sentiment. It is possible that the Karen topic was more negative than reported in the table, considering that the 2% positive, 79% neutral, and 11% negative posts in the Karen topic only account for a sum total of 92% of the sample. However, because the Karen topic constitutes only 0.1% of Aware's reported overall sample—and only 0.04% of ours—even if it were entirely negative, it would be impossible to shift the sample average by a magnitude of 3% assuming the average is unweighted. It is clear from the table of four example sentences that Aware does assign magnitudes to its sentiment and toxicity assessments, and so, it is unclear how Aware's data scientists translated these scores into the positive, negative, and neutral classificatory categories reported in Table 4. However, because the 3% figure appears to derive from the additive difference of classificatory categories in Table 4, it does not appear to be rooted in an analysis of the precise magnitudes generated by the Aware system.

To better understand the nature of these reported numbers, we compare them to our collection of models and our own manual annotations, which we report in Table 5.

Our manual inspection predictably revealed that many of the posts were difficult to classify along standardized sentiment and toxicity axes. Based on our reading of Aware's managerial objectives into the history of workplace sentiment analysis, we elected to classify as negative any post that described a negative workplace experience—in particular with a difficult customer—that might reasonably be hypothesized to contribute to employee burnout and attrition. This yielded significantly more negative examples than reported by any other model, which suggests that conflict between employees and customers is not primarily what Aware or any other model is identifying. We labeled as neutral posts that were either factual in nature, irrelevant, or uninterpretable without more context. We did not label any posts as positive, as of those that discussed frontline retail contexts using the term "Karen," none discussed positive customer experiences. That said, many posts framed negative experiences in positive language, either relating a past negative experience in a comedic tone or else expressing relief or support in light of such experiences. These posts may help to explain the positive sentiments detected by the other models, as the category of positive sentiments toward

Table 4 What % of messages are (Aware 2023)

What % of messages are...	Overall results	Karen topic	Difference
Positive	8	2	− 6%
Neutral	86	79	− 7%
Negative	8	11	+ 3%
Healthy	89	80	− 9%
Toxic	11	20	+ 9%

Table 5 Sentiment and toxicity model comparison on Karen dataset

	Aware	Llama-2	Vader	Detoxify	GPT	Manual
Positive	2%	10 (16%)	35 (55%)			0 (0%)
Neutral	79%	22 (34%)	8 (12%)			15 (23%)
Negative	11%	32 (50%)	21 (33%)			46 (77%)
Healthy	80%	32 (50%)		7 (11%)	41 (64%)	54 (84%)
Toxic	20%	32 (50%)		57 (89%)	23 (36%)	10 (16%)

those identified by commenters as Karens is otherwise difficult to explain.

Following Aware's use of toxicity to describe instances of harassment or other matters of personal employee culpability, we labeled as toxic instances in which posters were engaged in interpersonal disputes that might contribute to a hostile work environment, often involving labeling one another as "Karens." Notably, none of these instances involved discussions of customers. It is therefore unclear, when Aware discusses the rise in toxicity caused by Karens, whether it may be reading into the retail frontline context properties of the messages themselves as classified by its platform.

We re-ran our models on a modified dataset containing edited versions of each post to explore the factors contributing to sentiment and toxicity classifications. As we expected, replacing obscenity or high polarity words such as "grouchy" and "entitled" with neutral words or eliminating clauses containing those words was often able to decrease the negativity and toxicity of posts, even if the intent of the message still appeared clearly negative or toxic to human annotators. This suggests that these models remain fairly sensitive to surface-level features of messages, which is not surprising.

3.4 Discussion

In applying contemporary sentiment analysis technologies to the long standing question of managing worker sentiment, Aware blurs the line between divergent notions of sentiment. By replicating an experiment conducted by Aware data scientists to the extent possible given the publicly available information, we have created a context in which it is now possible to examine more closely how Aware may be adapting the notion of sentiment within its domain of application, and what potential risks or complications may arise in the computational analysis of workplace sentiment.

The fundamental problem in applying sentiment and toxicity analysis to workplace communications is that of defining sentiment and toxicity. Star ratings, likes, and other social media metadata provide an intuitive way to operationalize sentiment analysis as a problem of predicting the metadata from the text. Llama-2's reported ability to classify a significant proportion of Aware's internal data correctly, coupled with the areas of overlap between Aware's reported results and our own, suggests that Aware may be at least partially trained on extant sentiment analysis data, raising questions about whether these data reflect the type of managerial sentiment or toxicity that constitute Aware's central focus.

To the extent that Aware has specialized its platform to the employee communication domain, which lacks naturally occurring metadata such as star ratings, Aware researchers themselves, according to the report, supply such metadata

themselves by annotating datasets of employee communications. The process of annotation, in turn, raises questions about how annotators—and indeed managers themselves—can reliably determine the sentiment of an employee from a single Reddit comment. Aware hedges against the risk of algorithmic decision-making by noting that their system only brings potential issues to the attention of managers, who may then exercise their human judgment. However, the histories of the Hawthorne study and the BSR speak to the difficulties that even teams of social scientists observing employees around the clock under controlled laboratory conditions can face when it comes to the question of understanding the internal sentiments that drive employee behavior.

A key risk of the flexibility of sentiment and toxicity as concepts is that they leave much room in their interpretation for bias to creep in. The figure of the Karen is a highly recognizable gendered and racialized stereotype. Aware's definition of a Karen as "a customer who escalates complaints beyond what is considered reasonable" is gender neutral. However, this definition apparently paraphrases a sentence from Wikipedia, now deleted, but which was current around the time of the report's publication, defining a Karen as a, "white woman perceived as entitled or demanding beyond the scope of what is normal" ("Karen (Slang)" 2023). Moreover, this definition is accompanied, in Aware's report, with a stylized illustration of a blond, white woman (Aware 2023). This gendering is in spite of the fact that many usages of the term referred explicitly to men, often supplementing "Karen" with terms, such as "Ken," "Kevin," or "man-Kanren." A large majority of the comments used the term in a general, hypothetical way that obscured the gender of the referent, potentially in cases reflecting the commenters' own internalized assumptions about the genders and associated personalities of their customers. Still others coined terms such as "Karanagers" for managers who are "entitled and demanding," undermining the central thesis of Aware's report that customers were to blame for employee burnout.

Even so, the fraction of the 64 Karen comments that did seem to support Aware's thesis would seem barely significant in a dataset of 152,716 comments. The fact that this Karen topic became the central thesis of the report speaks to one of the most significant potential risks of algorithmic sentiment analysis platforms. Namely that, as was the case in this report, small mathematical or typographic errors can confirm existing gendered and racialized preconceptions and become freighted with additional significance through the authority of statistics and algorithmic technology.

Our results indicate that algorithmic judgements of sentiment and toxicity seem to hinge on textual markers that do not necessarily align with the historical goals of managerial sentiment analysis. Moreover, particularly in the hands of managers untrained in data science, numerical anomalies that may reinforce gendered and racialized stereotypes can

contribute to discriminatory workplace decision-making even if the platform itself, as documented by the third-party algorithmic audits, does not differentially discriminate on the basis of protected categories.

4 Conclusion

In this study, we coupled a critical analysis of the rhetoric surrounding sentiment analysis bossware systems with a data scientific investigation of one such system. As we have shown, “sentiment” and “toxicity” are multifaceted concepts, and the application of sentiment analysis and toxicity detection to workplace communications risks misrepresenting the intent behind those communications. More generally, we join existing scholarship in calling for more critical engagements with the underlying epistemologies on which these systems trade. While this study was enabled by the serendipitous “natural experiment” afforded by Aware’s publicizing of its report, the mounting pressure on companies to make public details of their systems may enable more such investigations in the future (International Panel on the Information Environment 2024b). However, such a future is far from certain. Even in the time it took to publish these results, Aware itself has been absorbed by human risk management platform Mimecast (Mimecast 2024), and the Aware website, including the report at the center of this study, has been taken offline. This study therefore represents a preliminary attempt to explore the types of methods and questions through which scholars may exercise additional oversight of these algorithmic systems.

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Data availability The Reddit data used in this study are publicly available (Heitmann 2023).

Declarations

Conflict of interest On behalf of all authors, the corresponding author states that there is no conflict of interest.

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